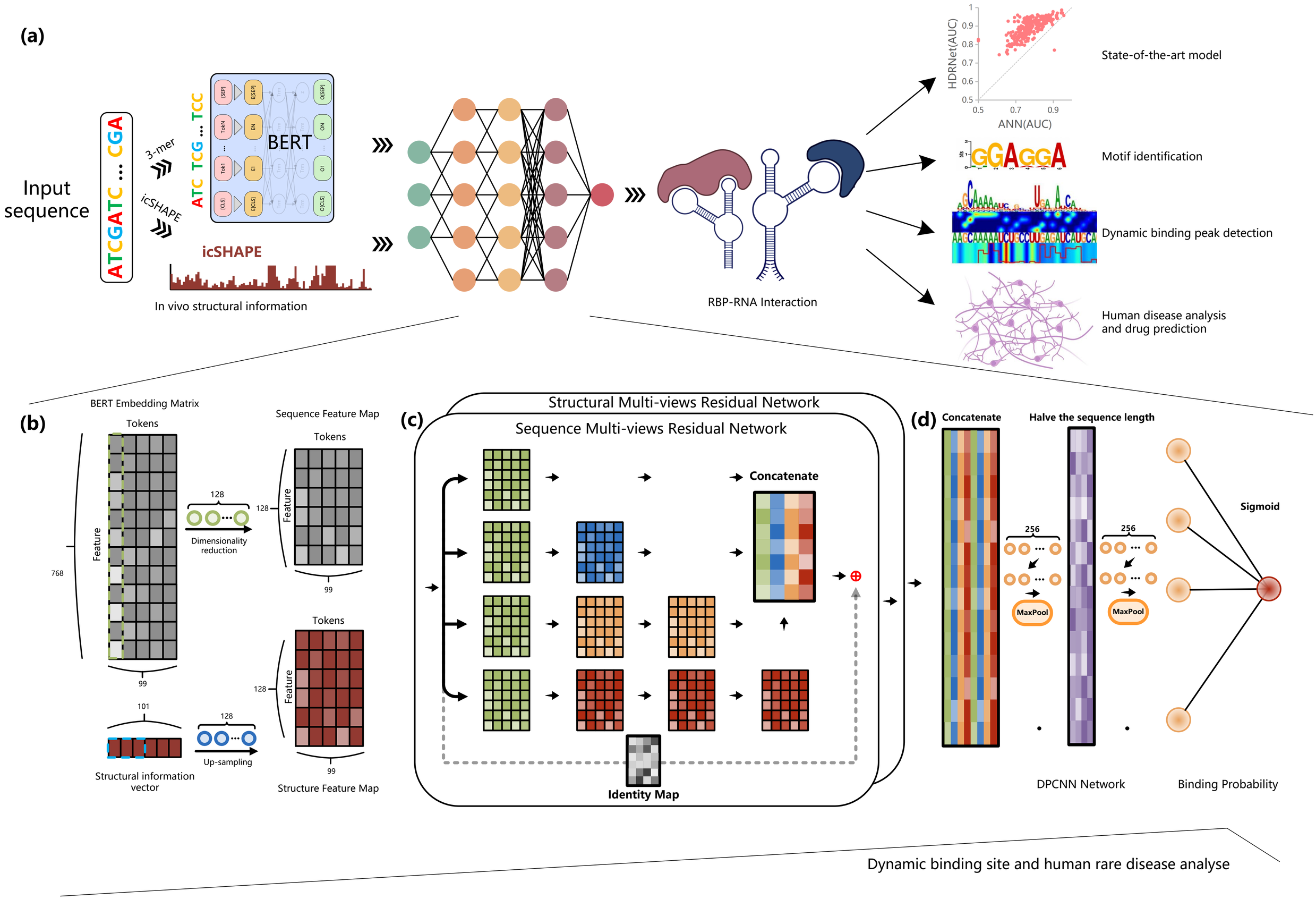
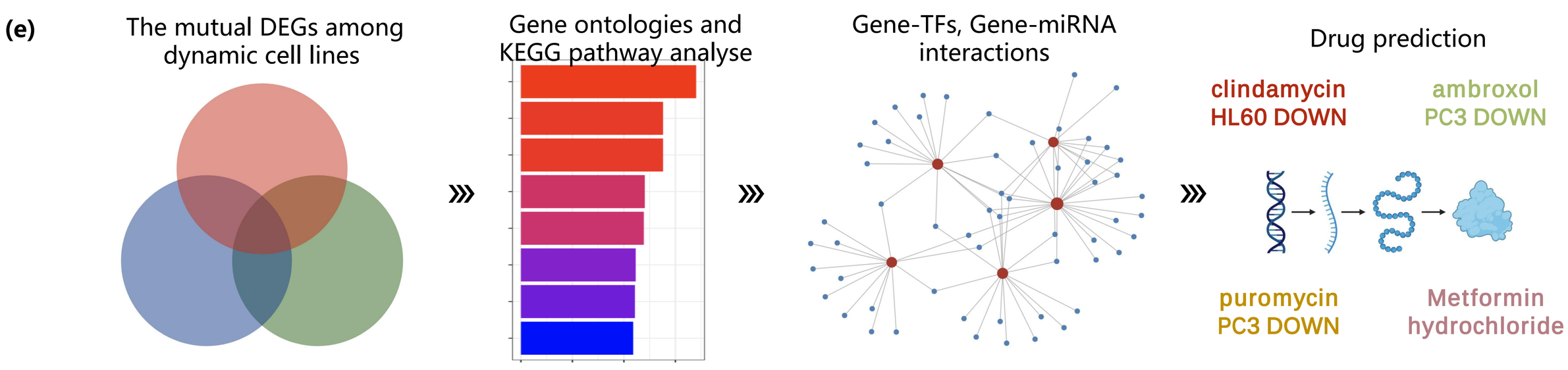

(a)
Input sequence
ATCGATC ... CGA
3-mer
ATC TCG ... TCC
BERT
icSHAPE
icSHAPE
In vivo structural information
RBP-RNA Interaction
HDRNet(AUC)
ANN(AUC)
State-of-the-art model
Motif identification
Dynamic binding peak detection
Human disease analysis and drug prediction
(b)
BERT Embedding Matrix
Tokens
Feature
768
99
128
Dimensionality reduction
Sequence Feature Map
Tokens
Feature
128
99
101
Structural information vector
Up-sampling
Tokens
Feature
128
99
Structure Feature Map
(c)
Structural Multi-views Residual Network
Sequence Multi-views Residual Network
Concatenate
Identity Map
(d)
Concatenate
Halve the sequence length
256
MaxPool
256
MaxPool
Sigmoid
DPCNN Network
Binding Probability
Dynamic binding site and human rare disease analyse


(e)
The mutual DEGs among dynamic cell lines
Gene ontologies and KEGG pathway analyse
Gene-TFs, Gene-miRNA interactions
Drug prediction
clindamycin HL60 DOWN
ambroxol PC3 DOWN
puromycin PC3 DOWN
Metformin hydrochloride